

Exercise 1. Show that $p^2 + 5$ is not a prime for any $p \geq 5$.

Exercise 2. For what n is the following integer divisible by 3?

$$1 - 2 + 3 - 4 + \cdots + (-1)^n n.$$

Exercise 3. Show that $a^6 \equiv 1 \pmod{7}$ for all integers a s.t. $7 \nmid a$.

Exercise 4. Show that last digit of 7^k and 7^l are same if $k \equiv l \pmod{4}$. Find the last digit of 7^{7^7} . Here, for computing the digits, the numbers are written in the usual fashion (base 10).

Exercise 5. Show that $1^n + 2^n + 3^n + 4^n$ is divisible 5 if and only if $n \not\equiv 0 \pmod{4}$.

Exercise 6. Show that the fraction $\frac{21n+4}{14n+3}$ is irreducible for every positive integer n .

Exercise 7. Show that $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Exercise 8. Find all integers r such that r can be written as a sum of ≥ 2 consecutive positive integers.

Exercise 9. Show that if $n \equiv 3 \pmod{4}$ then n cannot be a sum of two squares.

Exercise 10. Show that $p^6 + q^6 + 19$ is prime for at most one pair of primes (p, q) .

Bonus: Show that it is never a prime!